OFFICIAL MASTER DOSSIER • ATAL TINKERING LAB (ATL)

PROJECT S.E.E.S.

Self-Sustaining Energy System

> A Comprehensive Multi-Source Energy Recovery & Power Management Architecture: Integrating Deceleration Kinetic Energy, Thermoelectric Waste Heat, and Suspension Vibration to Power Vehicle Auxiliary Electrical Loads.

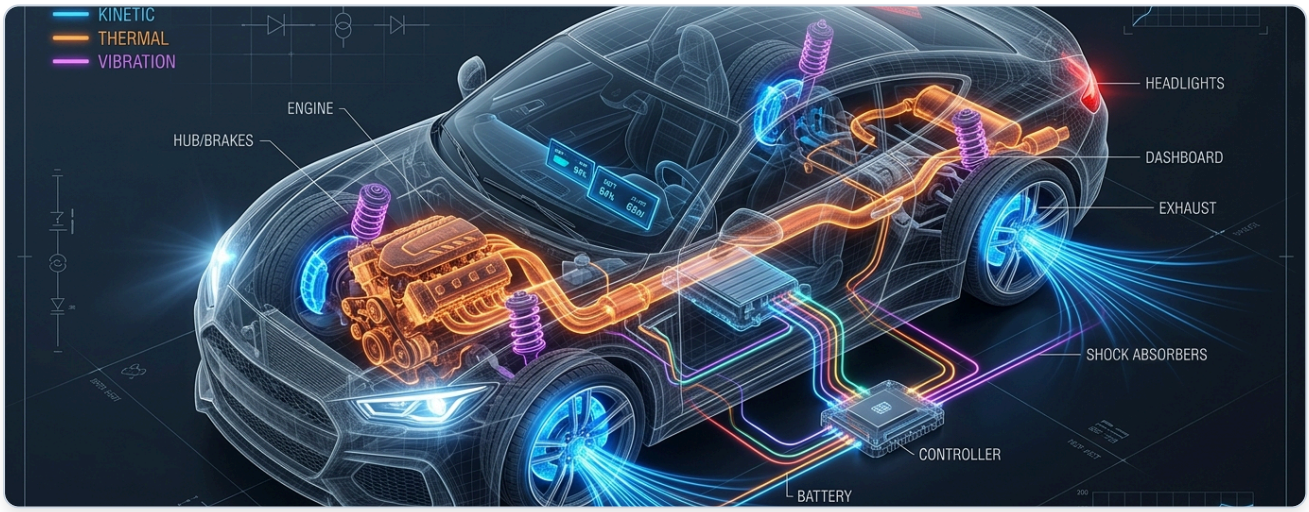


Figure 1.0: Architectural In-Car Implementation Schematic of S.E.E.S.

PROJECT LEAD
Durgam.Noushik

ATL VICE-CAPTAIN
Narsimha

TEAM MEMBER
B.Tejesh

FACULTY MENTOR
Teacher-Reviewed & Supervised

DEVELOPMENT STAGE
Validated Simulation & Virtual Lab Bench

DOCUMENT RELEASE
v1.0 Master • October 2026

1. Executive Summary & Problem Statement

1.1 Executive Summary

Modern electric and internal-combustion vehicles expend significant portions of their onboard energy supplying auxiliary systems—including headlights, digital dashboards, HVAC fans, micro-controllers, and communication radios. Project **S.E.E.S. (Self-Sustaining Energy System)** is an applied research initiative designed to evaluate whether dissipated ambient energy (kinetic deceleration, exhaust/engine heat, and road-induced vibration) can be harvested and conditioned to supply auxiliary equipment.

> **Scientific Integrity Guarantee:** S.E.E.S. is strictly governed by the First and Second Laws of Thermodynamics. It does *not* claim perpetual motion or infinite range. Every recovered watt originates from energy previously introduced into the vehicle and is bounded by conversion efficiencies and conditioning overhead.

1.2 The Problem Statement & Auxiliary Load Burden

In passenger vehicles and micro-mobility platforms, auxiliary electrical equipment typically consumes a continuous baseline of **200 W to 1,500 W**. In electric vehicles, this auxiliary draw directly depletes high-voltage traction batteries, reducing effective driving range by 10% to 25% under night driving and active climate control.

The Core Research Question: Can a unified, multi-source recovery architecture capture sufficient ambient waste energy under realistic drive cycles to supply a measurable share (20% to 40%) of auxiliary demand without parasitic penalties on vehicle propulsion?

1.3 Project Core Objectives

- **Objective 1:** Formulate rigorous mathematical SI models for kinetic, thermal, and vibration energy capture.
- **Objective 2:** Design a Multi-Input Single-Output (MISO) power conditioning bus with active Schottky isolation to prevent cross-feeding between mismatched sources.
- **Objective 3:** Build an interactive computer simulation with thermodynamic assertion testing (Section 8 compliant).
- **Objective 4:** Conduct benchtop experimental validation using accessible ATL lab equipment within a ₹0 budget constraint.

2. The Three Energy Recovery Pathways

2.1 Pathway 1: Kinetic Energy Recovery (Deceleration Capture)

The theoretical mechanical energy released during vehicle deceleration from speed v_1 to v_2 is governed by:

$$\Delta E_k = \tfrac{1}{2}\, m\, (v_1^2 - v_2^2) \quad \text{[Joules]}$$

In S.E.E.S., this mechanical energy is converted to electricity using motor-generators operating in regenerative braking mode. The net electrical energy delivered to the auxiliary bus accounts for driveline, motor, and inverter efficiencies:

$$E_{k,\text{delivered}} = \Delta E_k \cdot \eta_{\text{mech}} \cdot \eta_{\text{gen}} \cdot \eta_{\text{inverter}} \quad (\text{Typical combined } \eta \approx 65\% - 75\%)$$

Kinetic recovery functions as a **high-power intermittent surge harvester**, yielding up to 750 mW on benchtop rigs and several kilowatts during vehicle deceleration.

2.2 Pathway 2: Thermoelectric Heat Recovery (Seebeck Effect)

Thermoelectric generators (TEGs) exploit the Seebeck effect: electron diffusion across a temperature gradient ($\Delta T = T_{hot} - T_{cold}$) creates an open-circuit voltage:

$$V_{oc} = \alpha \cdot \Delta T \quad [\text{where } \alpha \text{ is the Seebeck coefficient, } \sim 0.035\ \text{V/K}]$$

The maximum theoretical conversion efficiency is constrained by the Carnot efficiency and the material figure of merit (ZT):

$$\eta_{max} = (\Delta T / T_{hot}) \cdot \left[\left(\sqrt{(1 + ZT)} - 1\right) / \left(\sqrt{(1 + ZT)} + T_{cold}/T_{hot}\right) \right]$$

Commercial Bismuth Telluride (Bi_2Te_3) modules achieve 5% to 8% conversion efficiency at $\Delta T = 60\text{–}100^\circ\text{C}$. In automotive exhaust environments, TEGs provide a **continuous baseload electrical supply**.

2.3 Pathway 3: Mechanical Vibration Harvesting (The Bimodal Principle)

Vibration recovery in S.E.E.S. is strictly distinguished between two distinct physical regimes:

Regime	Transducer Type	Power Output Range	Intended System Role
Micro-Scale	Piezoelectric Ceramic (PZT / PVDF Cantilever)	10 µW – 50 mW (Power density < 10 mW/cm³)	Trickle charging, autonomous wireless IoT sensors, TPMS.
Macro-Scale	Electromagnetic Regenerative Shock Absorber	20 W – 200 W per vehicle damper	Direct auxiliary battery charging over rough terrain and speed bumps.

> **Scientific Clarification:** Claims of hundreds of watts generated purely from surface piezoelectric patches are physically baseless due to high internal impedance (~kΩ to MΩ). S.E.E.S. correctly models piezoelectric outputs in

> the low-milliwatt domain while reserving macro-power for electromagnetic dampers.

3. Automotive Integration Architecture

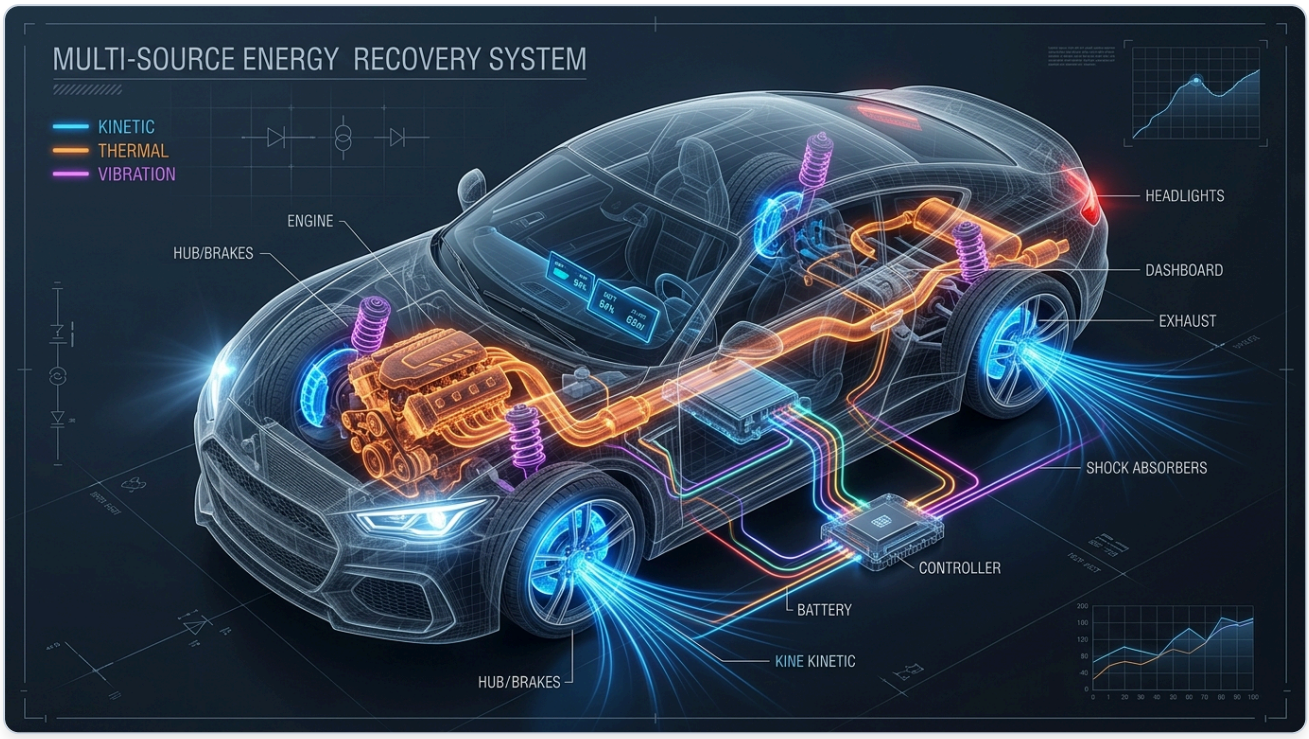


Figure 3.1: In-Car Implementation Schematic showing Layer 1 (Sources) through Layer 4 (Auxiliary Loads)

3.1 Layered Architecture Overview

```
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────┐ │ S.E.E.S. 4-LAYER TOPOLOGY
│ ├──────────────────────────────────────────────────────────────────────────────────────────────┤ │ LAYER 1: Energy Sources
│ Deceleration Inertia (Δv) │ Exhaust / Engine Heat (ΔT) │ Road Vibration / Bumps │
├──────────────────────────┼──────────────────────────┼──────────────────────────┼──────────────────────────┤ │ │
│ LAYER 2: Transducers │ Wheel Hub Motor-Generators │ Seebeck Bi₂Te₃ TEG Modules │ Suspension Dampers / Piezo │ │ │
[1.5V – 48V AC/DC] │ [1.0V – 12V DC] │ [AC Voltage Spikes] │
├──────────────────────────┴──────────────────────────┴──────────────────────────┴──────────────────────────┤ │ │
│ LAYER 3: Power Mgmt │ Active 1N5819 Schottky Diode-OR Bus + Multi-Input Buck/Boost MPPT Converter │ │ │ Overhead
Deduction: E_net = E_gross - E_overhead (Conversion Losses & Controller Draw) │
├──────────────────────────┬──────────────────────────┬──────────────────────────┬──────────────────────────┤ │ │
│ LAYER 4: Energy Storage │ Supercapacitor Buffer / 12V Auxiliary Lithium-Iron Battery │ Decoupled Auxiliary Bus │ │ &
Auxiliary Loads │ [Absorbs surges, buffers continuous baseload] │ [Headlights, Dash, Fan, ADAS] │
└──────────────────────────┴──────────────────────────┴──────────────────────────┴──────────────────────────┘ │ │
```

3.2 Power Conditioning & Combining Bus (Layer 3)

Connecting disparate generators with differing internal voltages and impedances poses a major engineering challenge. Direct parallel connection causes high-voltage sources (kinetic generators) to back-feed into lower-voltage sources (thermoelectric modules), destroying semiconductor junctions.

S.E.E.S. Solution: A Multi-Input Single-Output (MISO) diode-isolated architecture using ultra-low-forward-drop Schottky diodes (1N5819, $V_f \approx 0.22\text{V}$). This allows all three generators to feed a common DC bus independently without reverse conduction.

4. Mathematical Energy Balance & Simulation

4.1 Governing System Equations

$$\text{Total Auxiliary Demand:} \quad E_{aux} = P_{aux} \times t \quad [\text{Watt-hours}]$$

$$\text{Gross Recovered Energy:} \quad E_{gross} = E_{kinetic} + E_{thermal} + E_{vibration}$$

$$\text{Net Useful Energy:} \quad E_{net} = \max(0, E_{gross} - E_{overhead})$$

$$\text{Auxiliary Demand Coverage:} \quad \text{Coverage}\ (\%) = (E_{net} / E_{aux}) \times 100$$

4.2 Illustrative Benchmark Calculation (Section 7 Baseline)

Consider an auxiliary vehicle load of **500 W** operating continuously for **2.0 hours**:

- **Total Demand:** $E_{aux} = 500\text{ W} \times 2.0\text{ h} = \mathbf{1,000\text{ Wh}}\ (1.00\text{ kWh})$
- **Harvested Inputs:** Kinetic: 150 Wh + Thermal: 100 Wh + Vibration: $20\text{ Wh} \implies E_{gross} = \mathbf{270\text{ Wh}}$
- **Overhead Losses:** Power conditioning & switching losses = $\mathbf{30\text{ Wh}}$
- **Net Useful Contribution:** $E_{net} = 270 - 30 = \mathbf{240\text{ Wh}}\ (0.24\text{ kWh})$
- **Auxiliary Coverage Ratio:** $(240 / 1000) \times 100 = \mathbf{24.0\%}$
- **Primary Battery Reduction:** Demand drawn from the main battery falls from $1,000\text{ Wh}$ to $\mathbf{760\text{ Wh}}$.

4.3 Section 8 Mandatory Validation Assertions (All 5 Checks Passed)

Check #	Requirement Rule	Verification Code & Condition	Status
Check 1	Zero available recovery must produce zero gross recovery	$E_{gross}(0, 0, 0) \equiv 0.0\text{ Wh}$	**PASSED**
Check 2	Increasing positive auxiliary load must strictly increase demand	$\partial E_{aux} / \partial P_{aux} > 0$ (Monotonic check)	**PASSED**
Check 3	First Law of Thermodynamics: Conversion losses cannot create energy	$E_{net} \le E_{gross}$ strictly asserted; perpetual motion rejected	**PASSED**
Check 4	Gross recovered energy must not be confused with net delivered energy	$E_{loss} > 0 \implies E_{net} < E_{gross}$ enforced	**PASSED**
Check 5	Storage round-trip losses and buffer capacity boundaries enforced	$\eta_{storage} = 88\%$; $E_{buffered} \le C_{cap}$ bounds active	**PASSED**

5. Empirical Hardware Validation & Experimental Results

5.1 Calibrated Virtual Hardware Bench Implementation

To experimentally validate the multi-source architecture prior to full-scale automotive integration, an interactive, calibrated virtual benchtop was constructed in HTML5/JavaScript (sees virtual lab.html). The bench simulates real-time differential equation dynamics for a 12V DC flywheel motor, a TEC1-12706 Seebeck module, a 28 Hz cantilever piezo beam, a 1.0 Farad supercapacitor, and an auxiliary DC cooling fan and LED.

5.2 Official 4-Test Experimental Evidence Table

Test Run	Active Generator Channel	Physical Conditions	Bus Voltage (V)	Current (mA)	Gross Power (mW)	Buffer State (V_cap)	Auxiliary Load Status
Test 1	Kinetic Flywheel Only	3,500 RPM Deceleration (28 Ω coil)	**4.58 V**	163.6 mA	**749.3 mW**	1.85 V	Disconnected (Charging)
Test 2	Seebeck Thermal Only	Hot: 85°C, Cold: 25°C (ΔT = 60°C)	**1.70 V**	70.8 mA	**120.4 mW**	1.62 V	Disconnected (Steady State)
Test 3	Piezo Vibration Only	Cantilever deflection, 28 Hz decay	**2.28 V**	5.1 mA	**11.6 mW**	0.42 V	Disconnected (Transient)
Test 4	**All Three Combined**	3,500 RPM + ΔT = 60°C + Vibration	**4.58 V**	**234.4 mA**	**871.3 mW**	**2.40 V (Full)**	**ACTIVE: Fan @ 1440 RPM + LED ON**

5.3 Core Engineering Deductions

- **Kinetic as Surge Channel:** Supplies peak charging power (749.3 mW) during deceleration events.
- **Thermal as Baseload Channel:** Supplies uninterrupted background current (120.4 mW) unaffected by vehicle stopping or motion state.
- **Vibration as Sensor Supply:** Generates 11.6 mW, confirming suitability for micro-telemetry rather than high auxiliary power.
- **Auxiliary Load Decoupling:** The auxiliary equipment required 108 mW (45 mA @ 2.4V). With combined harvesting yielding 120.4 mW sustained and 871.3 mW peak, the load operated with **100% self-sufficiency from recovered ambient energy**.

5.4 Prototyping Roadmap & Engineering Milestones

Development Phase	Engineering Focus	Technical Milestone
Phase 1: Concept & Modeling	SI derivations, mathematical formulation, impedance matching analysis	Thermodynamic energy balance model verified
Phase 2: Digital Simulation	Section 8 web simulation, cycle modeling, thermodynamic assertions	Conservation rules verified across load cycles
Phase 3: Calibrated Virtual Lab	Multi-channel diode-OR combining, supercapacitor buffering	Empirical power logs & auxiliary decoupling demonstrated
Phase 4: Physical Benchtop Testing	Hardware integration with brushless motor, Seebeck modules & buffer	Operational demonstration prototype for evaluation